# Chapter 2

# Terminology

## In this chapter…

- **Find definitions for common terms used in the BenMAP-CE tool and in this manual.**

**Active Layer.** In the GIS window, the active layer is the top-most data layer. All queries or statistical analysis of the map will act upon this top-most layer.

**Aggregation.** The summing of grid cell level results to a larger spatial scale, such as county, state, or national levels.

**Aggregation, Pooling, and Valuation (APV) Configuration.** APV configurations store your preferences regarding how to aggregate your results, whether and how to pool your results, and any economic valuation functions you have applied. For example, an APV file might aggregate your estimated change in incidence to the U.S. county level, it might pool across multiple hospital admission health impact functions and it could include an economic valuation function. APV configurations are stored in files with an *.apvx* file extension. The results derived from an APV configuration have an *.apv**r**x* file extension. APV files are by default stored in the *APV* folder, and APV results files are by default stored in the *CFGR* folder.

**Air Quality Surface.** An air quality surface contains modeled or monitored air pollution data in a series of cells; these cells may be a regular shape (like a 12km by 12km grid) or an irregular shape (like a county or census tract). These surfaces are also referred to as air quality grids. BenMAP-CE uses one air quality grid to represent the baseline scenario and a second grid to represent the control scenario. These baseline and control grids must share the same geographic structure. The program calculates the difference between baseline and control grids as an input to the health impact function. Air Quality Grids are stored in files with an *.aqgx* file extension.

**Air Quality Metric.** The metric expresses the time period over which air quality values are modeled or observed and whether that modeled or observed air quality value is an average, maximum or minimum. For example, the metric DailyMean represents the average concentration for the sampling day. This could be taken directly from a single 24-hour observation or from an average of hourly (or more frequent) observations. In addition to the time period, some metrics also specify the method used for averaging or aggregation. For example, a typical ozone metric D8HourMax represents the highest of the 8-hour moving averages during the day.

**Air Quality Model.** Air quality models are valuable air quality management tools. Models are mathematical descriptions of pollution transport, dispersion, and related physical and chemical processes in the atmosphere. Air quality models (like CMAQ[9] and CAMx[10]) are used to estimate the air pollutant concentration at specific locations, which are referred to as receptors, or over a spatial area that has been divided into uniform grid squares. The number of receptors or grid-cells in a model far exceeds the number of monitors one could typically afford to deploy in a monitoring study. Therefore, models provide a cost-effective way to analyze pollutant impacts over a wide spatial

[9] Community Multi-scale Air Quality (CMAQ) Model is available at: http://www.epa.gov/amad/Research/RIA/cmaq.html or https://www.cmascenter.org/cmaq/.

[10] Comprehensive Air Quality Model with Extensions (CAMx) is available at: http://www.camx.com/.

area where factors such as meteorology, topography, and emissions from both local and remote sources could be important. BenMAP-CE does not contain an air quality model.

**Attainment.** The state of meeting the National Ambient Air Quality Standard (NAAQS) for a pollutant. A geographical area that meets the NAAQS is called an "attainment area."

**Audit Trail.** This is a report that contains a record of all the choices involved in creating a particular file. Audit trails can be created for any file that BenMAP-CE creates.

**Background Concentration.** The concentration of a pollutant, generally in the absence of human sources.

**Background Incidence.** The incidence of a given adverse effect due to all causes including air pollution. Also called baseline incidence rates.

**Baseline Scenario.** The air quality levels prior to whatever policy change you are evaluating. The baseline is sometimes referred to as “Business as Usual.” The baseline scenario is usually considered to be the reference scenario against which to compare a potential “control scenario”, in which air quality levels are changed from the baseline levels.

**Beta.** The coefficient for the health impact function. The value of beta (ß) typically represents the percent change in a given adverse health impact per unit of pollution.

**Closest Monitor.** The procedure by which data from the closest monitor is used to represent air pollutant levels in a population grid cell. BenMAP-CE can also scale the data from the closest monitor with air pollution modeling data. BenMAP-CE includes two types of scaling - “temporal” and “spatial”. See “Scaling” for additional information.

**Community Multi-scale Air Quality (CMAQ) Model.** An open-source photochemical grid air quality model that the U.S. EPA and others rely upon to predict levels and changes in pollutant concentrations.

**Concentration-Response (C-R) Function.** A C-R function estimates the relationship between adverse health effects and ambient air pollution, and is used to derive health impact functions (defined below). You will often see that the term C-R function and health impact function are used interchangeably.

**Configuration.** A Configuration stores the health impact functions and model options used to estimate adverse health effects. Configurations are stored in files with a *.cfgx* file extension. CFGX files are by default stored in the *CFG* folder. The results derived from a Configuration have a *.cfgrx* file extension. CFGR files are by default stored in the *CFGR* folder.

**Contingent Valuation.** A survey-based economic technique for the valuation of non-market resources, such as environmental preservation or avoidance of air pollution health risk.

**Control Scenario.** In a modeling study, this is a sensitivity scenario in which emissions from one or more source sectors are changed (increased or decreased) from a given "baseline scenario". The control scenario generally represents air quality levels after a new policy has been implemented.

**Core BenMAP-CE.** The fully-featured benefits analysis program that accepts user-defined air quality data, quantifies health impacts, aggregates, values and pools results (details available in Section 3.1). Non-Core BenMAP-CE features include Command Line and PopSim, described in the following Chapters and Appendices.

**Cost of Illness (COI).** The cost of illness includes the direct medical costs and lost earnings associated with illness. These estimates generally understate the true economic value of reductions in risk of a health effect, as they include just the direct expenditures related to treatment and lost earnings but not the value of avoided pain and suffering.

**Currency Year.** The value of the currency based on the year specified. Valuation estimates should use a consistent currency year to account for inflation. For example, you might want to report the valuation estimate in 2000 dollars to make it easier to compare with your cost analysis, which uses that same currency year.

**Deltas.** The difference between two data points. As used in BenMAP-CE, mapping the air quality deltas shows the change in air pollution between the baseline air quality grid and the control air quality grid.

**Discount Rate.** In a cost-benefit analysis, the discount rate is a quantitative method to account for the fact that people generally value future benefits and costs less than current costs and benefits. Typically, if a benefit occurs over multiple years, the economic benefit would be discounted.

**Endpoint.** An endpoint is a subset of an endpoint group, and represents a more specific class of adverse health effects. For example, within the endpoint group *Mortality*, there might be the endpoints *Mortality, Long Term, All Cause* and *Mortality, Long Term, Cardiopulmonary*.

**Endpoint Group.** An endpoint group represents a broad class of adverse health effects, such as premature mortality or hospital admissions. BenMAP-CE only allows pooling of adverse health effects to occur within a given endpoint group, as it generally does not make sense to sum the number of cases of disparate health effects, such as premature mortality and hospital admissions.

**Epidemiology.** The study of factors affecting the health and illness of populations. Epidemiological studies cannot prove that a specific risk factor actually causes the disease being studied but can only show that a risk factor is associated (correlated) with a higher incidence of disease in the population exposed to that risk factor.